

FuzzyShield

Real-Time Behavioral Malware Classification
Using Fuzzy Logic

Term Project Presentation

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Outline

- 1 Introduction
- 2 Malware Profiles
- 3 System Design
- 4 Asymmetric EWMA
- 5 Implementation
- 6 Results
- 7 Conclusion

The Problem: Signatures Are Not Enough

Signature-Based Detection

- Matches binary hashes / byte patterns
- Requires a **known sample** to exist first
- Fails on **zero-day** & polymorphic malware
- Recompiling the binary evades detection

Behavioral Analysis

- Observes *what a process does*
- CPU, disk I/O, network, entropy
- Works on **unknown** samples
- Complements signatures

Zero-Day Reality

Average exploit-to-patch window: **70 days**.
Signature databases are **useless** during this window.

Key Insight

Ransomware always **encrypts files**.
Trojans always **exfiltrate data**.
Miners always **max out CPU**.
Behaviour is hard to hide.

Why Fuzzy Logic?

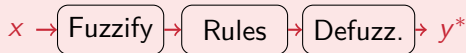
Behavioral metrics are inherently imprecise:

- CPU = 78% — normal *or* malicious?
- fwrite = 85 MB/s — high *or* extreme?
- Hard thresholds produce false positives

Fuzzy sets handle the grey zones:

- 78% CPU $\Rightarrow$ high(0.60), extreme(0.20)
- Smooth transitions, not hard cuts
- Human-readable IF-THEN rules
- **Transparent** — analyst can inspect every rule

Mamdani FIS Pipeline



$$y^* = \frac{\int y \mu_{\text{agg}}(y) dy}{\int \mu_{\text{agg}}(y) dy}$$

Centroid defuzzification

Three Malware Families — Behavioral Fingerprints

Type	CPU	Disk W	Net TX	Conns	Primary Behavioral Signal
Ransomware	High	Very High	Silent	Few	fwrite + entropy ≈ 8 + many extensions
Trojan	Low	Minimal	Very High	Many	nettx + conn count + priv escalation
Cryptominer	Extreme	None	Moderate	Few	cpu=extreme + fwrite ≈ 0 + pool traffic
Benign	Idle	Minimal	Silent	None	All metrics below detection threshold

Ransomware

Encrypts files with AES/ChaCha20.
Demands ransom for decryption key.
Output entropy ≈ 8 bits/byte —
indistinguishable from random data.

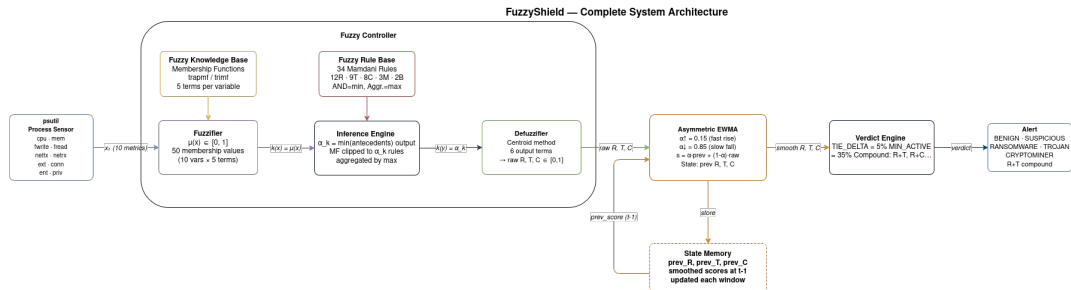
Trojan / Backdoor

Disguises as legitimate process.
Exfiltrates data to C2 server.
Maintains many persistent
outbound connections.

Cryptominer

Proof-of-work on CPU (SHA-256).
Mines Monero (XMR) via pool.
Near-zero disk I/O.
Low-entropy stratum protocol.

FuzzyShield Architecture



Input Layer

10 behavioral metrics
sampled via `psutil`

Inference Engine

34 Mamdani rules
min-AND, max-aggregation

Output Layer

3 threat scores
centroid defuzzification

10 Behavioral Input Variables

Variable	Range	Terms (5)
cpu	0–100%	idle ... extreme
mem	0–2048 MB	tiny ... huge
fwrite	0–200 MB/s	none ... extreme
file_read	0–200 MB/s	none ... extreme
nettx	0–10000 KB/s	silent ... flood

Variable	Range	Terms (5)
netrx	0–10000 KB/s	silent ... flood
ext	0–50	single ... mass
conn	0–500	none ... swarm
ent	0–8 bits/B	ordered ... max
priv	0–20	none ... extreme

Ransomware signals

fwrite, ext, ent

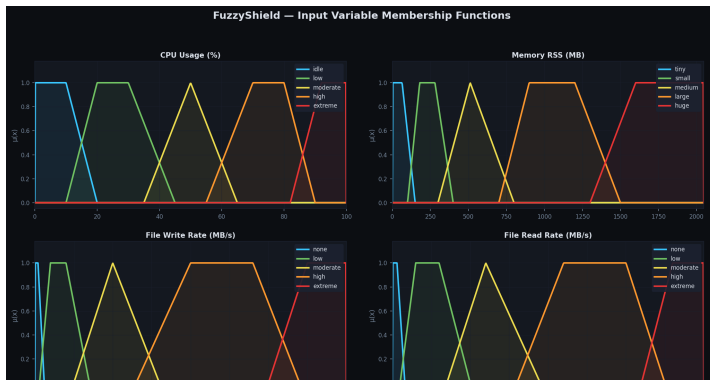
Trojan signals

nettx, conn, priv

Cryptominer signals

cpu, mem, fwrite $\approx$ 0

Membership Functions — All 10 Inputs



- **Trapezoidal** (trapmf) and **triangular** (trimf) MFs — overlapping boundaries for smooth transitions
- `cpu[extreme]`: starts at 82% — real miners saturate *all* cores
- `ent[max]`: ≥ 7.2 bits/B — indistinguishable from AES output
- `conn[few]`: 3–50 conns — pool (≈ 8) vs trojan swarms (300+)

*Showing cpu, mem, fwrite, file_read.
All 10 MFs in report Fig. 2.*

Shannon Entropy — Key to Ransomware Detection

Definition

$$H = - \sum_{b=0}^{255} p_b \log_2 p_b \quad [\text{bits/byte}]$$

Data type	H (bits/byte)
-----------	-----------------

Plain text (.txt)	≈ 4.5
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Executable (.exe)	≈ 6.0
-------------------	---------------

ZIP / compressed	≈ 7.5
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AES ciphertext	≈ 7.9
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Measurement method

Reads **last 4 KB** of open writable files, computes H on byte value frequencies.

ent [max]: $H \geq 7.2$

Computationally indistinguishable from random data at this level.

High fwrite + ent [max]

⇒ **Ransomware CRITICAL**

Miner contrast: pool uses

ent [low] ($H < 4.5$) — structured.

Output Variables & Verdict Generation

3 Output Channels ($\in [0, 1]$)

Term	Score range
none	$< 15\%$
trace	$5\% - 35\%$
low	$25\% - 55\%$
medium	$45\% - 75\%$
high	$65\% - 92\%$
critical	$> 85\%$

Centroid: $y^* = \frac{\int y \mu_{\text{agg}} dy}{\int \mu_{\text{agg}} dy}$

Compound Verdict Algorithm

- 1 $\max(R, T, C) < 0.10 \Rightarrow$ **BENIGN**
- 2 $\max < 0.40 \Rightarrow$ **SUSPICIOUS**
- 3 Collect threats: $\text{score} \geq \max - 0.05$
AND $\text{score} \geq 0.35$
 $\Rightarrow$ join as **R + T**
- 4 $\max \geq 0.70 \Rightarrow$ **HIGH CONFIDENCE**

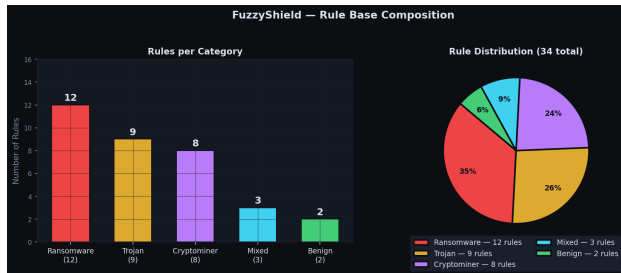
Compound Example

$R = 50.6\%$, $T = 51.0\%$

Both $\geq 35\%$, within $\Delta = 5\%$

$\Rightarrow$ **RANSOMWARE + TROJAN**

Rule Base — 34 Mamdani Rules



Rule structure

IF *cond* AND *cond*...

THEN *output* = *term*

Firing strength (min T-norm):

$$\alpha_k = \min_i \mu_{A_{ik}}(x_i)$$

- **12** Ransomware — fwrite, ext, ent
- **9** Trojan — nettx, conn, priv
- **8** Cryptominer — cpu, mem
- **3** Mixed / ambiguous
- **2** Benign suppression

Key Rules — Ransomware

#	Antecedent	Output
R1	<i>fwrite=extreme</i> & <i>ext=many</i> & <i>ent=max</i>	R= critical
R2	<i>fwrite=high</i> & <i>ext=many</i> & <i>ent=high</i>	R= high
R6	<i>fwrite=high</i> & <i>ent=max</i> & <i>priv=mod</i>	R= high
R25	<i>fwrite=high</i> & <i>ext=several</i> & <i>ent=max</i>	R= high
R27	<i>fwrite=high</i> & <i>ent=max</i> & <i>conn=none</i>	R=medium
R28	<i>fwrite=extreme</i> & <i>fread=low</i> & <i>ent=max</i>	R= critical

Design decisions

R1 — strongest signal:

All three dominants active at once
(write extreme + mass extensions
+ max entropy) = encrypt-in-place

R27 — offline case:

No C2 connection during encryption
⇒ medium score, not critical

R28 — overwrite pattern:

Read little, write extreme + max ent
= overwrite without reading first

Key Rules — Trojan & Cryptominer

Trojan rules

#	Antecedent	Out
R8	nettx= <i>high</i> & conn= <i>many</i> & priv= <i>high</i>	crit.
R9	nettx= <i>flood</i> & conn= <i>swarm</i>	crit.
R11	cpu= <i>low</i> & conn= <i>many</i> & priv= <i>high</i>	high
R14	nettx= <i>flood</i> & priv= <i>extreme</i>	crit.
R30	fread= <i>low</i> & nettx= <i>high</i> & conn= <i>many</i>	high

Cryptominer rules

#	Antecedent	Out
R16	cpu= <i>extreme</i> & fwrite= <i>none</i> & nettx= <i>mod</i>	crit.
R17	cpu= <i>extreme</i> & mem= <i>large</i> & nettx= <i>mod</i>	crit.
R19	cpu= <i>extreme</i> & fwrite= <i>none</i> & conn= <i>few</i> & ent= <i>low</i>	high
R22	cpu= <i>extreme</i> & fwrite= <i>none</i> & conn= <i>few</i>	high
R24	cpu= <i>extreme</i> & fwrite= <i>none</i> & ent= <i>ordered</i>	crit.

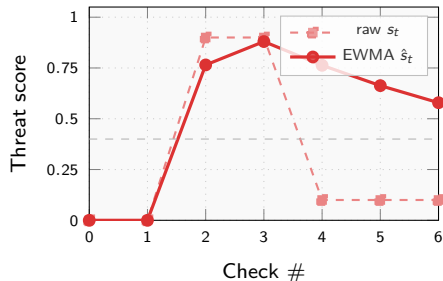
conn[few] prevents false positives from CPU-intensive non-miners.

Asymmetric EWMA — Temporal Feedback

Update Rule (per check)

$$\hat{s}_t = \alpha \hat{s}_{t-1} + (1-\alpha) s_t$$
$$\alpha = \begin{cases} \alpha_{\uparrow} = 0.15 & s_t \geq \hat{s}_{t-1} \\ \alpha_{\downarrow} = 0.85 & s_t < \hat{s}_{t-1} \end{cases}$$

- **Rising** ($\alpha=0.15$): new value gets **85%** weight
⇒ threat alarm raised **in one window**
- **Falling** ($\alpha=0.85$): old score keeps **85%**
⇒ idle pause cannot reset alarm
- Defeats **burst-pause evasion**:
malware cannot hide by going briefly idle
- Verdict always on $\hat{s}_t$, not raw s_t



- Checks 2–3: burst → score rises fast
- Check 4: process idles → score stays above 40%
- *Without EWMA: check 4 would drop to 10%*

Dual Implementation: Python & MATLAB

Python — scikit-fuzzy

- skfuzzy.control Mamdani API
- psutil live PID monitoring
- /proc/pid/io wchar write rate
- Timestamped: scenarios/, pid_sessions/, analysis/
- Behavioral simulator: sim/malware_sim.py

MATLAB — Fuzzy Logic Toolbox

- mamfis API, addRule
- **Exact mirror** of 34 Python rules
- Exports FuzzyShield.fis (Fuzzy Logic Designer)
- 3D surface plots via evalfis
- Identical MF parameters $\Rightarrow$ same outputs

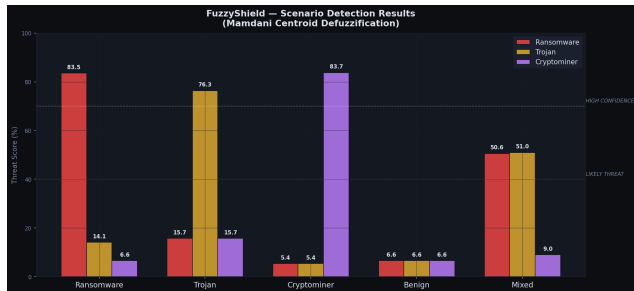
Behavioral Simulator — sim/malware_sim.py

Ransomware	8 workers $\times$ 500 KB os.urandom(), 20 ext., held 40 ms
Trojan	250 loopback TCP conns, echo server, $\approx$ 25 MB/s TX/RX
Cryptominer	N_{cores} SHA-256 threads + 8 pool connections ($\approx$ 640 KB/s)
Benign	idle sleep loop, minimal resource use

Preset Scenario Results

Scenario	R%	T%	C%
Ransomware	83.5	14.1	6.6
Trojan	15.7	76.3	15.7
Cryptominer	5.4	5.4	83.7
Benign	6.6	6.6	6.6
Mixed	50.6	51.0	9.0

- All 4 single threats $\geq 70\%$:
HIGH CONFIDENCE
- Benign $< 10\%$: **SAFE**
- Mixed: **R** + **T** compound



Why Each Scenario Fires

Ransomware — 83.5%

fwrite 160 MB/s (*extreme*)

ext 45 (*mass*)

ent 7.8 (*max*)

Rules fired: R1, R2, R5, R25, R28

Trojan — 76.3%

nettx 4200 KB/s (*high*)

conn 320 (*many*)

priv 14 (*high*)

Rules fired: R8, R9, R10, R12

Cryptominer — 83.7%

cpu 96% (*extreme*)

mem 1400 MB (*huge*)

fwrite 2 MB/s (*none*)

conn 20 (*few*)

Rules fired: R16, R17, R20, R22

Benign — 6.6%

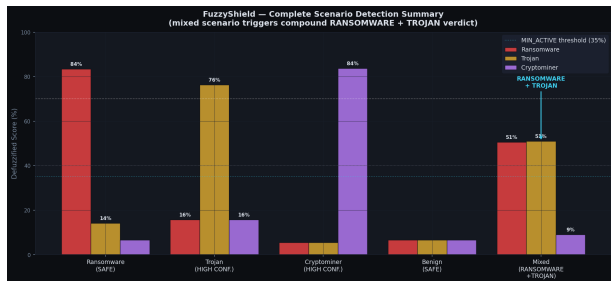
cpu 12% (*idle*)

fwrite 4 MB/s (*none*)

conn 4 (*none*)

Rules B1, B2 → all outputs suppressed

Compound Threat Detection



Mixed Scenario Inputs

fwrite 170 MB/s (*extreme*)
ent 7.8 (*max*)
ext 38 (*many*)
nettx 1800 KB/s (*high*)
conn 180 (*many*)
priv 10 (*high*)

Verdict

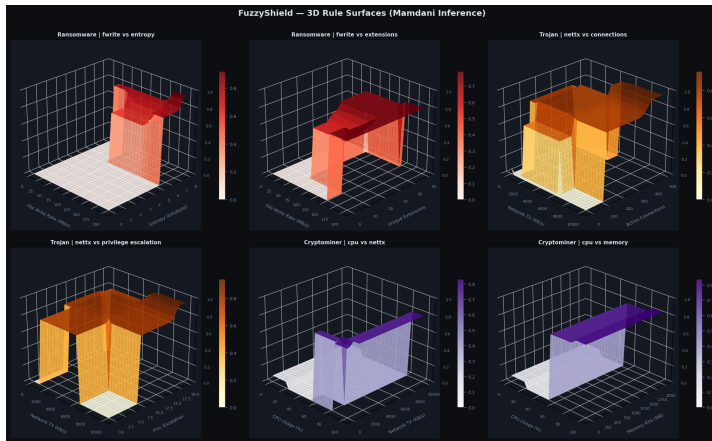
R = **50.6%**, T = **51.0%**

Both $\geq 35\%$, $\Delta < 5\%$

⇒ **RANSOMWARE + TROJAN**
LIKELY THREAT

Rule M3: priv + ent + conn fires both channels.

3D Rule Surfaces (MATLAB Surface Viewer)



Ransomware:

Sharp ridge — fwrite & ent
must *both* exceed “high”

Trojan:

Plateau at high nettx
& many connections

Cryptominer:

Narrow ridge: cpu=extreme
+ nettx=moderate (pool)

Conclusion

What was built

- Mamdani FIS: **10 inputs, 3 outputs, 34 rules**
- Real-time psutil-based PID monitor
- Compound verdict (multi-threat)
- Asymmetric EWMA ($\alpha_{\uparrow}=0.15$, $\alpha_{\downarrow}=0.85$)
- Python + MATLAB dual implementations

Detection accuracy

Threat	Score	Verdict
Ransomware	83.5%	HIGH CONF.
Trojan	76.3%	HIGH CONF.
Cryptominer	83.7%	HIGH CONF.
Benign	6.6%	SAFE

Limitations

- Web servers can mimic miner CPU profile
- Single-window snapshot → FP risk
- Async write buffers may delay entropy

Future Work

- Adaptive MF tuning via RL
- Trend-based temporal features
- eBPF kernel integration
- Wider taxonomy: worms, rootkits

Thank You

Questions?



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<https://github.com/Farukesgin/FuzzyShield>
FuzzyShield — Real-Time Behavioral Malware Classification
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